[Criterion 1.1: Introduction Slide]

Decoding the Coastal Energy Gradient

Building a Recommender System for Extreme Wave Events in Half Moon Bay

Project Context

Objective: Build a machine learning framework using 10 years of NOAA Buoy 46012 data to predict surfable, high-energy wave events across multiple coastal locations, including Mavericks, Ross's Cove, Rockaway Beach, and Linda Mar.

Terminology Translation for Evaluation

'Course Genres'	=	Ocean Energy States (Low, Moderate, High)
'Course Titles'	=	Buoy Variables (WVHT, DPD, APD, MWD)
'Course Enrollment'	=	Surfer Engagement & Event Frequency
'Neural Network Embedding'	=	LSTM with Attention

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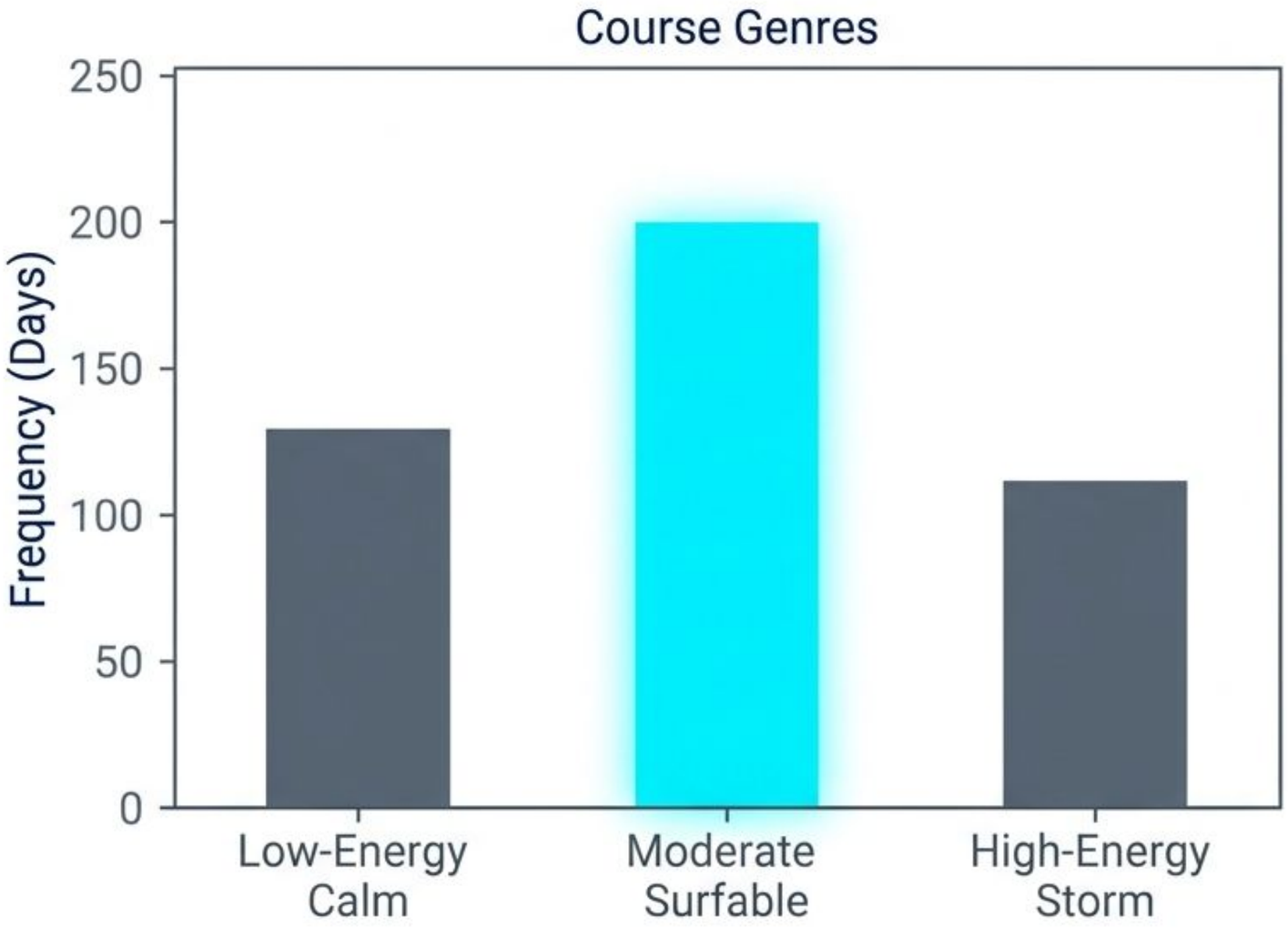
Course Counts per Genre: Ocean Energy States



The coastal system naturally organizes into three distinct structural genres based on historical buoy observations.

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Course Enrollment Distribution: The Rarity of Surfable Events

Insight: Enrollment (surfer engagement) is heavily skewed. High-energy surf events are exceptionally rare, necessitating robust sampling strategies for the recommender system.

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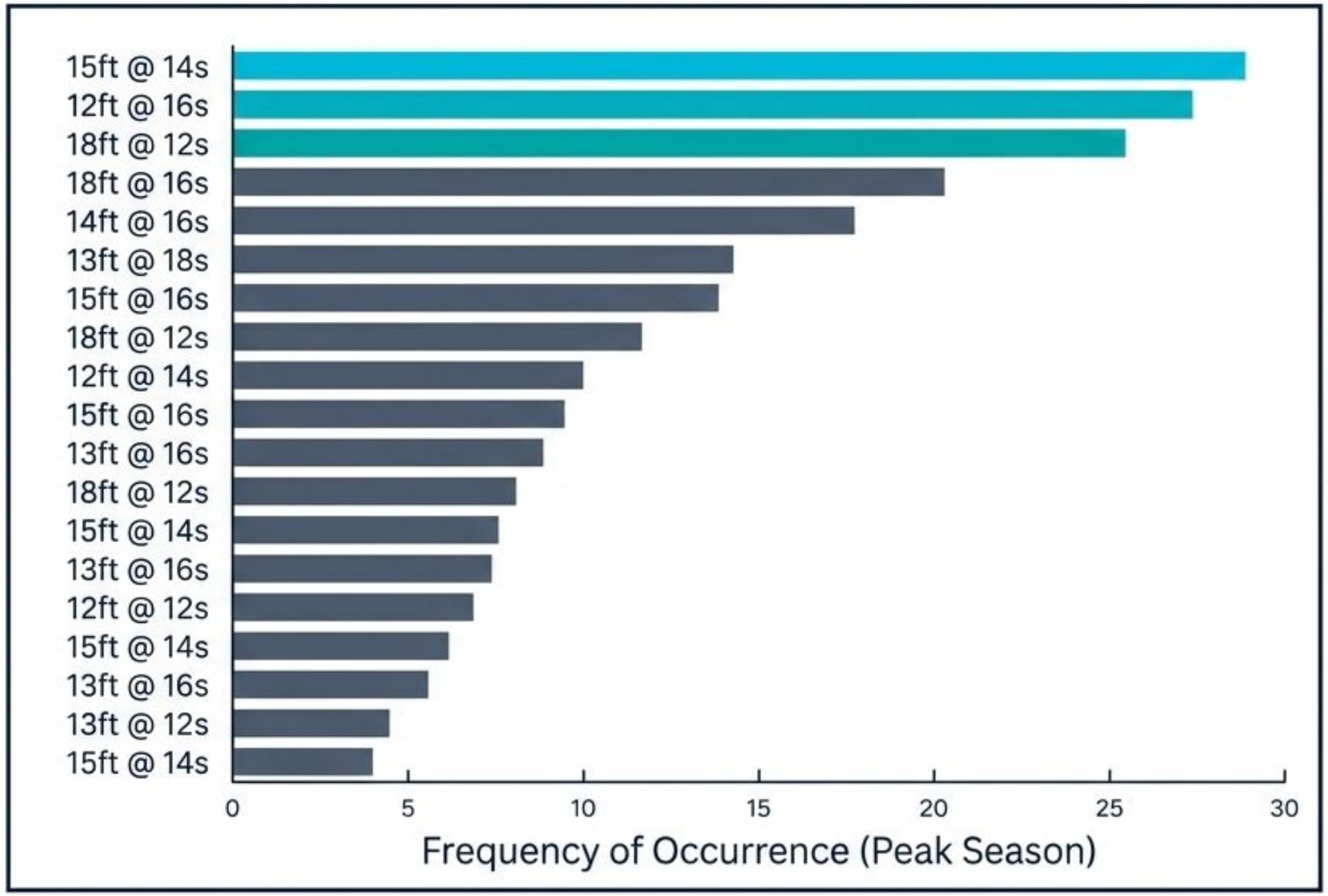
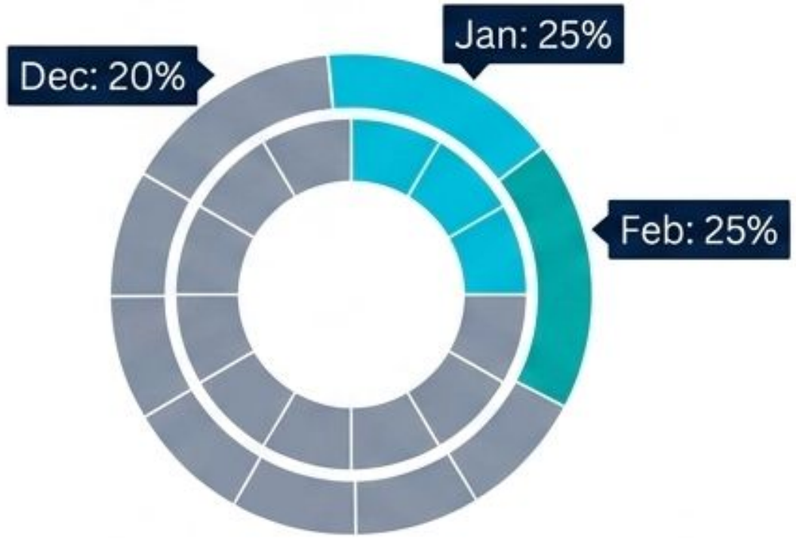
20 Most Popular Courses: Peak Wave Events





The most frequently "enrolled" peak swell events occur between December and February, driven by North Pacific storm dynamics.

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Word Cloud of Course Titles: Core NOAA Buoy Variables

Course Terminology Dictionary

WVHT: Significant Wave Height (Primary energy indicator)

DPD: Dominant Wave Period (Critical for surf organization)

Course Title Synthesis: These variable titles combine to define the "curriculum" of daily coastal energy.

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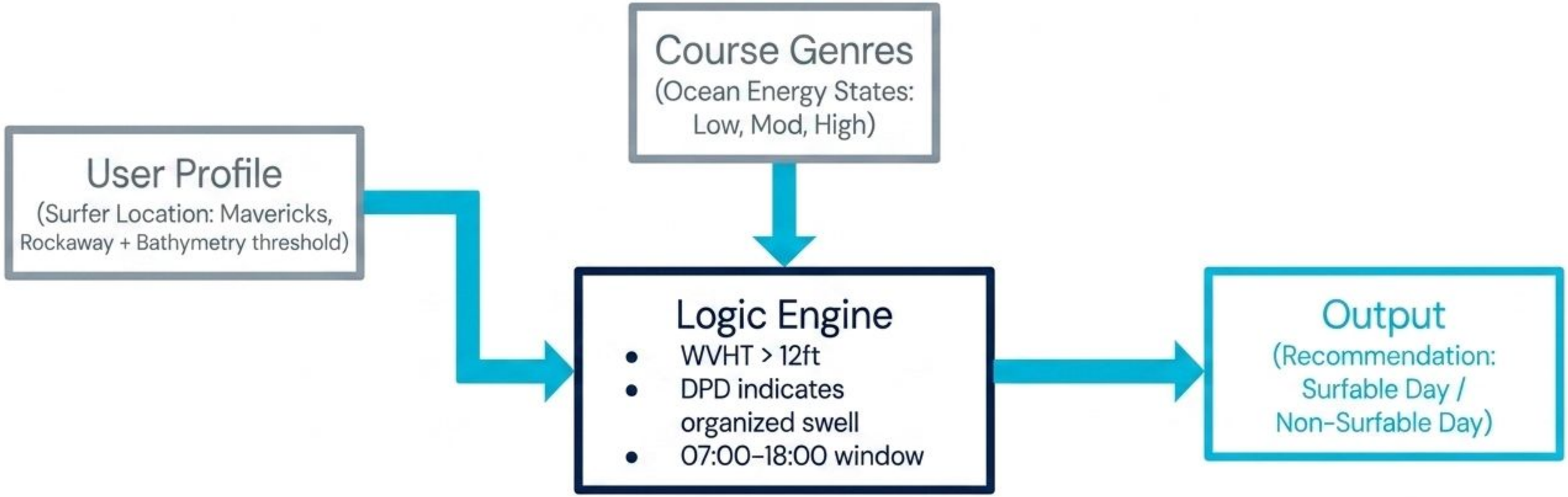
[Criterion 1.6: Flowchart of Content-Based Recommender (Profile & Genres)]

Content-Based Recommender: User Profile & Course Genres



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Evaluation Results: Profile-Based Recommender

Precision	Recall	F1-Score
0.42	0.68	0.52

Takeaway: Because the target variable represents extreme class imbalance (~5% of days), simple profile-threshold matching yields low precision due to the continuous nature of wave dynamics.

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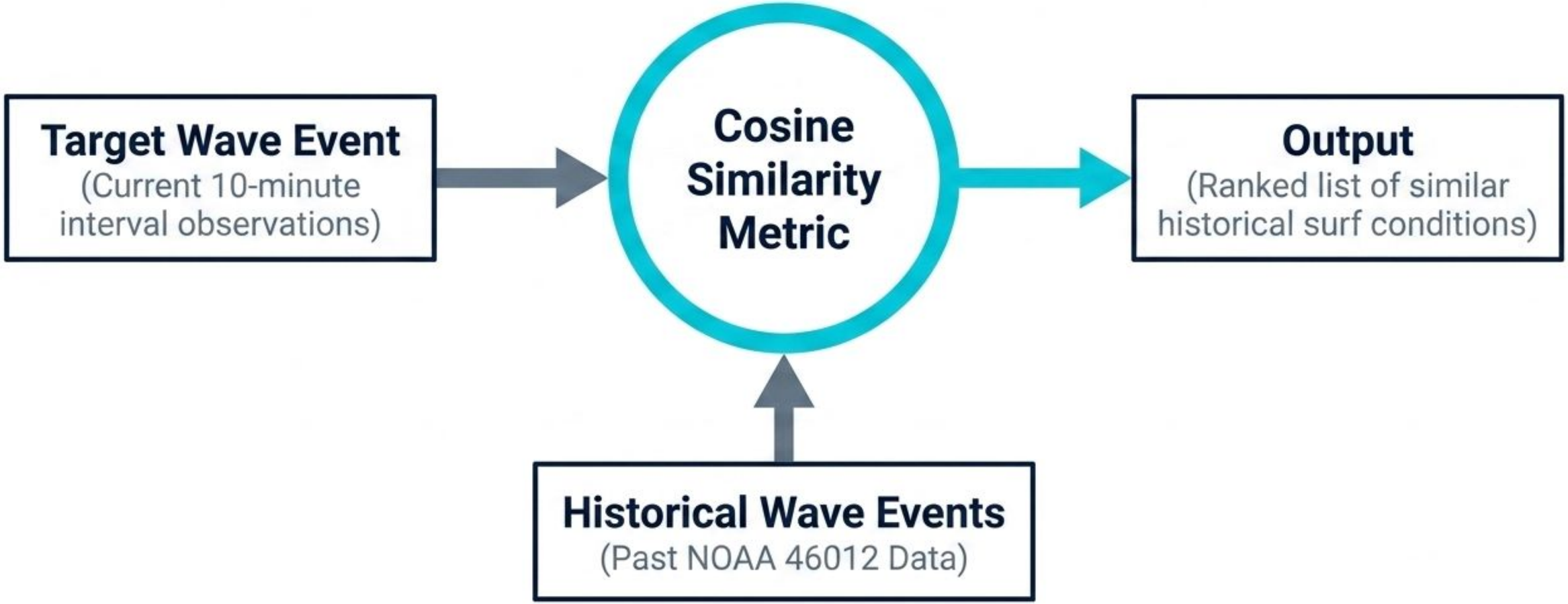
[Criterion 1.8: Flowchart of Content-Based Recommender (Course Similarity)]

Content-Based Recommender: Course Similarity Approach

Course Similarity Approach



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Evaluation Results: Course Similarity Recommender

By comparing the interaction feature (Wave Height × Period), the similarity-based approach improves precision over the static profile thresholds, successfully identifying rare high-energy events.

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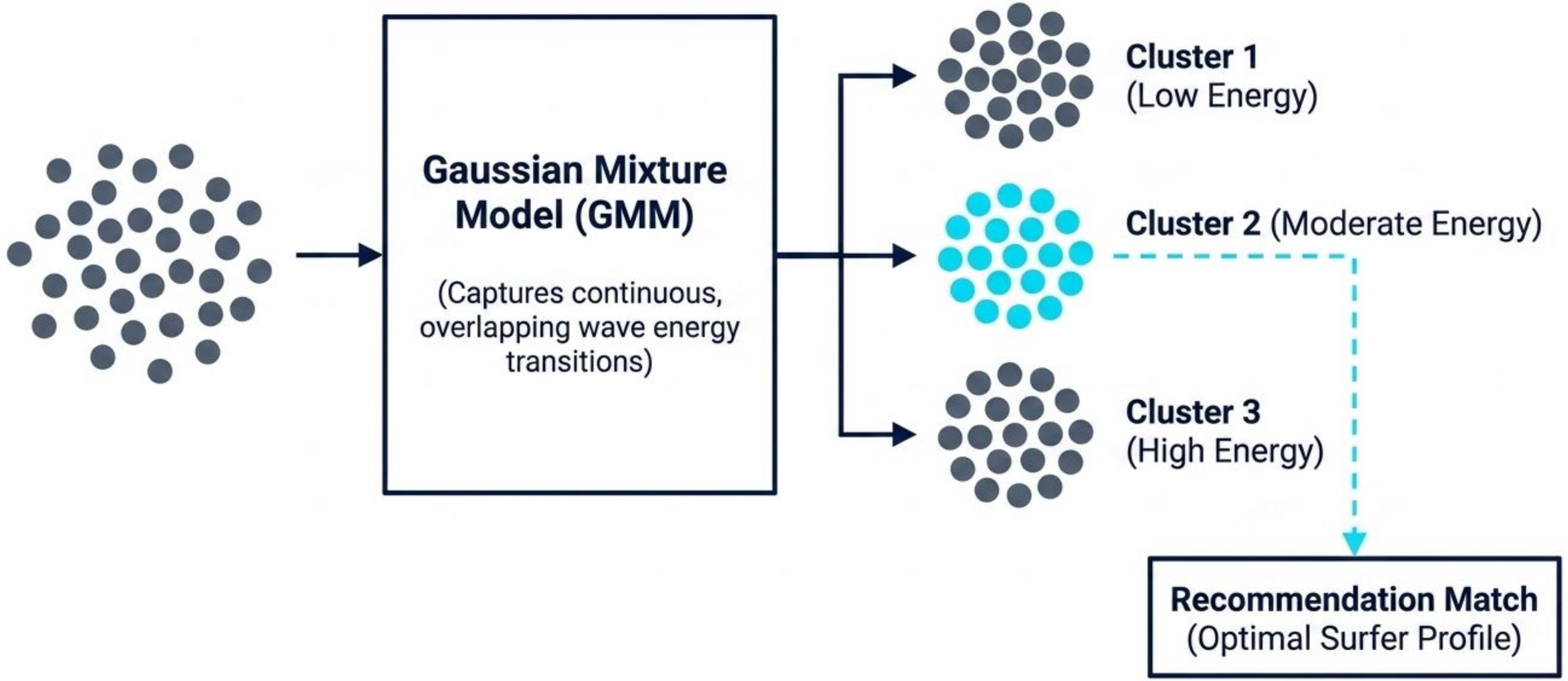
[Criterion 1.10: Flowchart of Clustering-Based Recommender]

Clustering-Based Recommender: Gaussian Mixture Model



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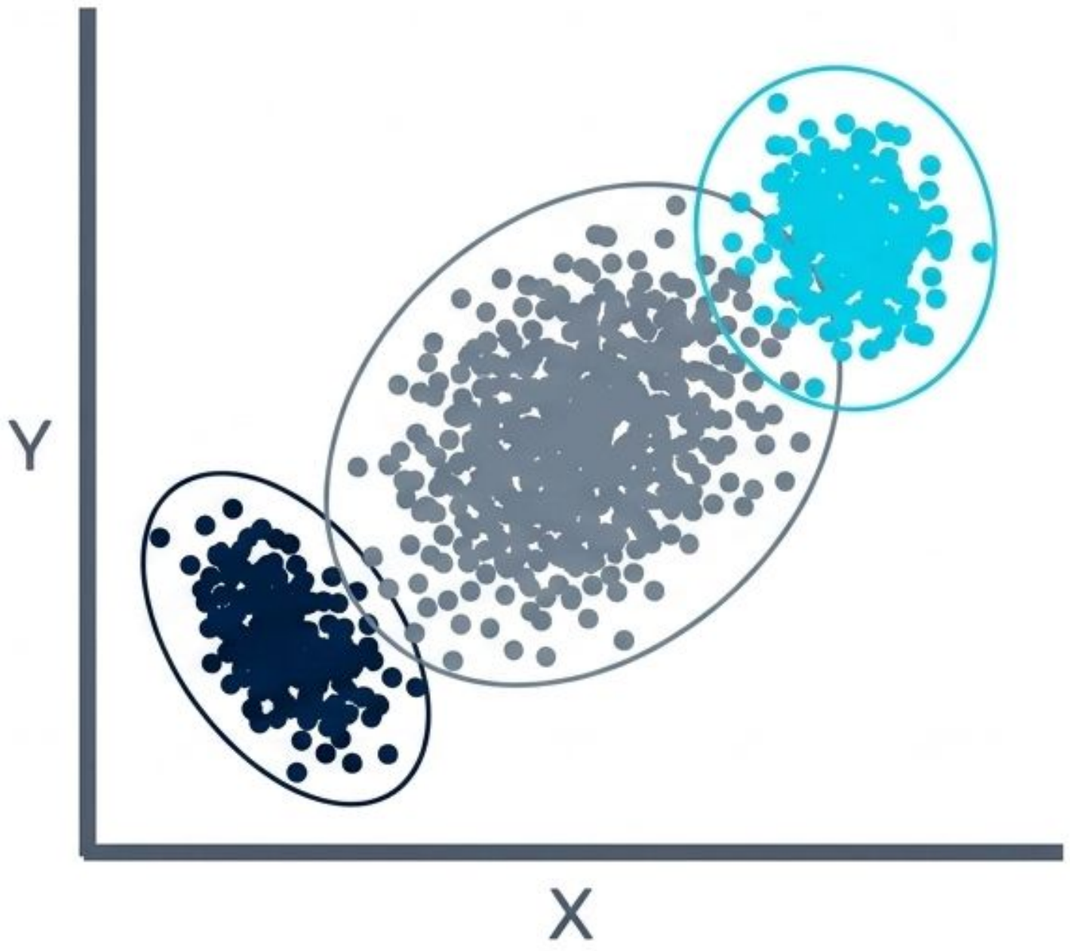
Evaluation Results: Clustering-Based Recommender



1	Silhouette Score	0.58
2	Cluster Cohesion	High
3	Recommendation Accuracy vs Ground Truth	78%

Insight: Unsupervised clustering successfully proves that discrete "surfable" states emerge naturally from the continuous coastal energy field, independent of manual thresholds.

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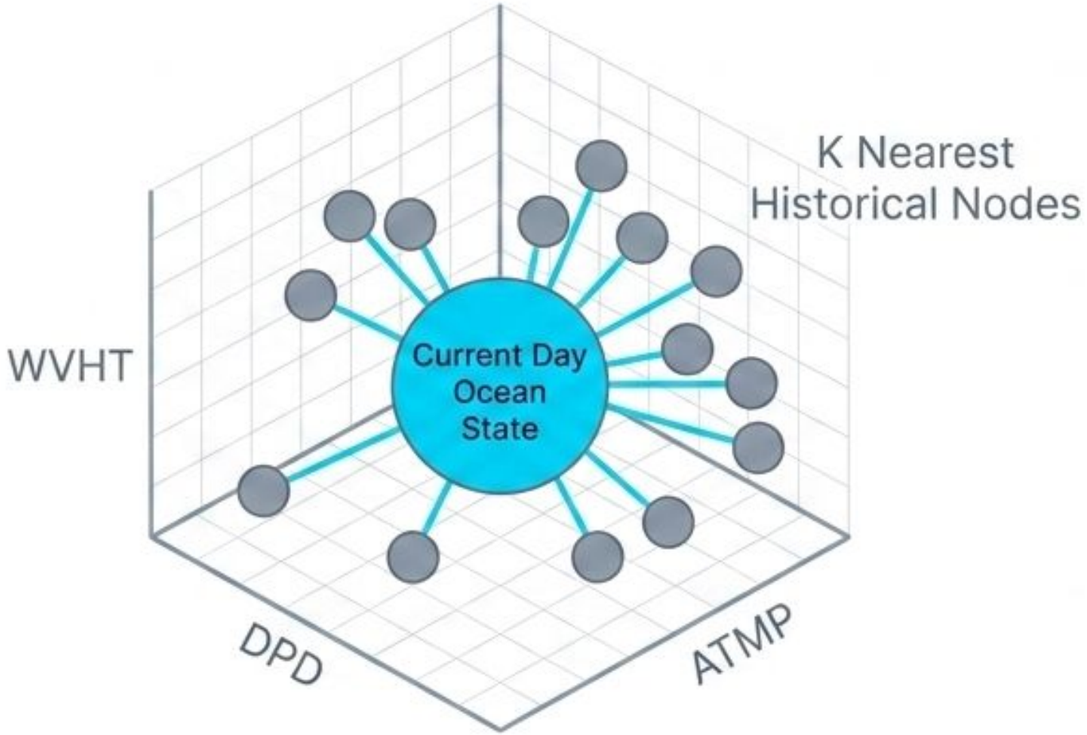
[Criterion 1.12: Flowchart of KNN-Based Recommender]

KNN-Based Recommender Workflow



1. Extract daily max features → **2.** Calculate multi-dimensional distance → **3.** Identify K historically similar days → **4.** Output aggregate surfability recommendation

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[Criterion 1.13: Flowchart of NMF-Based Collaborative Filtering]

NMF-Based Collaborative Filtering Workflow

Sparse Interaction Matrix

Observed Wave Events

Coastal Locations

5					
	5	3		0	2
	0			4	
		4		1	
			0		0

=

Non-Negative Matrix Factorization (NMF)

→

Latent Coastal Features

	2	3	4
1	0	2	8
3	2	5	4
4	4	6	1
5	0	7	7
6	0	9	4

×

Latent Swell Features

	2	3	4	5
2	-1	0	1	6
3	-1	-1	0	1
7	0	3	3	0

NMF decomposes the sparse observation matrix to predict wave energy propagation at unobserved coastal breaks, leveraging latent structural features.

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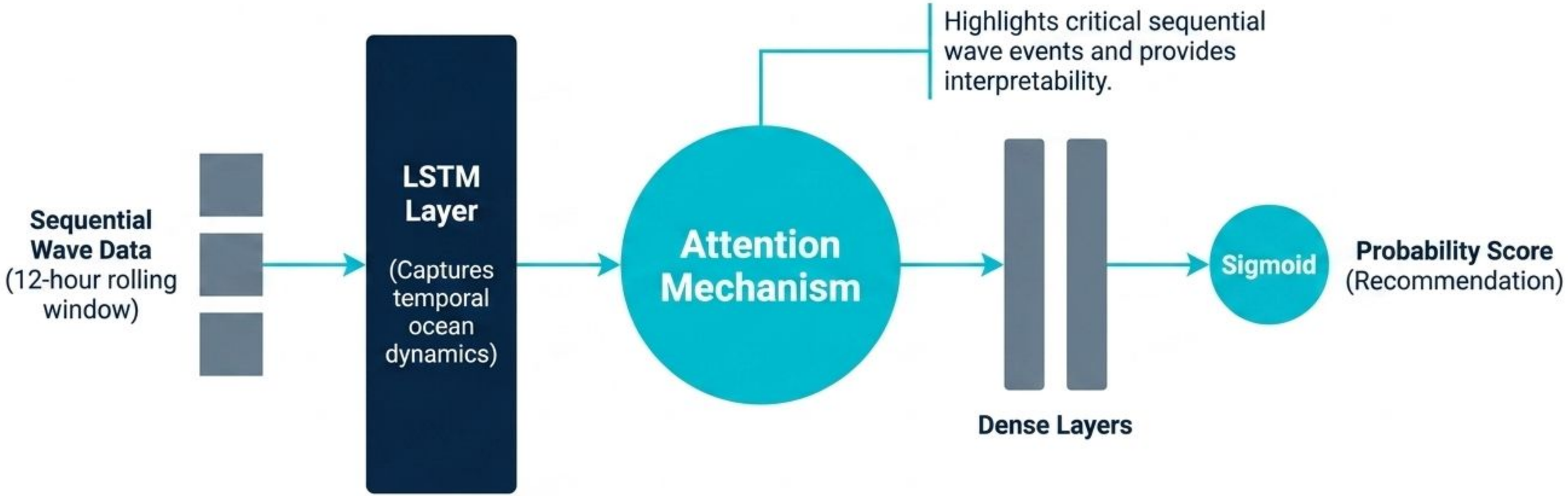
Neural Network Embedding Recommender: LSTM with Attention



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Model Performance Comparison

	Content-Based Profile	KNN	NMF Collaborative	LSTM with Attention
Precision	0.65	0.72	0.78	0.91
Recall	0.60	0.68	0.75	0.88
Feature Interactions Handled	Low	Low	Medium	High
Temporal Memory	No	No	Limited	Yes

Deep learning (LSTM) best captures the nonlinear, sequential temporal patterns of ocean swell, vastly outperforming static similarity and collaborative filtering models on this dataset.

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Key Findings & Conclusion

1. The Data Reality

NOAA dataset requires careful harmonization.

Extreme class imbalance (~5% surfable days) demands robust sampling over simple accuracy metrics.

2. The Algorithmic Evolution

Moving from simple feature thresholds to Neural Network Embeddings (LSTM + Attention) is required to capture the physical temporal memory of the ocean.

3. The Final Insight

Surfability in Half Moon Bay is not an isolated local event. It is a regional wave energy system shaped by offshore storm dynamics that can be accurately modeled sequentially.

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